Week 12 Project Presentation Guidelines (Stream 1)

MATH3888 (Interdisciplinary maths project unit)

Submission Deadline is **Friday, Week 12 (October 30th), 23:59**. No late submission will be accepted!

Submission is via Canvas. One group member submits:

- **pdf**-file of the corresponding presentation slides. It is expected that you use LaTeX (e.g., via Overleaf) or another professional software to format and create your slides.
- file name should be `groupXX-week12.pdf` where `XX` is your two-digit group number (e.g., group07-week12.pdf).

This group presentation is worth 30% of your final mark.

INSTRUCTIONS

Part 1: Format and Presentation Logistics

- The presentation will be a live slide presentation in your assigned classroom during Week 12.
- All group members must be present and contribute to the presentation.
- **Strict 16-Minute Block:** Your group has a strict 16-minute timeslot. This includes a maximum **8-minute presentation** followed by a **8-minute "lightning round"** panel Q&A and transition time.
- **Slide Constraints ($1 + N + 1$ Rule):** The presentation must be structured in landscape orientation with exactly 1 Title Slide, N Content Slides (where N is your number of group members), and 1 References Slide.
  - **1 Title Slide:** Including the project title, names (+SID) of all group members, and a clear mapping of exactly who is responsible for presenting which content slide.
  - N **Content Slides:** Every student is responsible for presenting exactly one content slide.
    (Note: For the 5-person group, some members may present two slides to cover all required content).
  - **1 References Slide:** For all literature and sources cited.
- **Cohesion and Consistency:** While each student is responsible for their own slide, the final presentation must flow as a single, unified scientific narrative. Ensure consistent formatting, terminology, and visual design across the entire slide deck. A fragmented or randomly mixed deck will negatively impact the baseline group mark.
- **Time Limit:** Each student has strictly **1 minute** to speak. You will be cut off if you exceed your limit.

Part 2: Q&A Lightning Round

Following the presentation, the grading panel will ask one targeted question to each group member based specifically on the slide they presented. You will have **45-60 seconds** to provide a concise answer.

Part 3: Suggested Slide Content Layout (8 Core Themes)

Your presentation must cover the 8 core areas below, which map directly to your Week 5 and Week 8 project pitch.

- **Slide 1: Biological Setting:** Briefly introduce the cell signaling system, physiological features, and a visual diagram of the model.
- **Slide 2: Experimental Protocols:** Detail the specific data, agonist/blocker interventions, and key literature graphs/tables.
- **Slide 3: Master Equations:** Present the ODE system and define the key variables.
- **Slide 4: Connecting Biology and Mathematics:** Explain how key biological components and parameters map to the flux terms in your master equations.
- **Slide 5: Baseline Dynamics:** Provide Python time-series plots of all variables alongside experimental time traces to validate baseline dynamics.
- **Slide 6: Bifurcation Analysis:** Show `PyDSTool` bifurcation diagrams mapping the dynamical changes against the agonist/stimulus.
- **Slide 7: Biological Interpretation:** Synthesize the results. How do the mathematical findings align with your original biological hypotheses?
- **Slide 8: Limitations & Conclusion:** Discuss any technical bottlenecks encountered and conclude with the main takeaways.

Note: 8-person groups map 1:1 to these themes with exactly 8 content slides. 9-person groups should assign two presenters to split a dense theme (like baseline dynamics) across their 9 content slides. The 7-person and 5-person groups must strictly maintain their N content slides (7 and 5, respectively) by logically merging some of the 8 themes onto a single slide.

Part 4: The 3Cs of Scientific Presentation

Part of your task is to determine how best to present the information and your scientific argument. Consider the 3Cs:

- **Considered:** Have you considered all aspects of the task?
- **Constructed:** Are your slides well-constructed, visually appealing, and easy to follow? Have you used diagrams and figures appropriately?
- **Concise:** Being concise brings clarity. Is the text on your slides accurate and easy to read?

Part 5: Assessment and Grading

This group presentation is worth 30% of your final mark. The grading will be split into two components:

- **Baseline Group Mark:** A collective score awarded for the overall quality, cohesion, and visual construction of the slide deck, as well as the successful synthesis of the interdisciplinary concepts across the whole presentation.
- **Individual Adjustment:** Your final individual mark may be adjusted up or down from the baseline group mark. This adjustment will be based entirely on your individual oral delivery and your ability to accurately and concisely answer the targeted question during the Q&A lightning round.

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